Sherwin Vahidimowlavi

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EDUCATION

Northeastern University, San Jose, CA — Sept. 2024 – Present
Khoury College of Computer Sciences — Expected Graduation: Dec. 2026
Candidate for Master of Science in Data Science
Related Courses: Deep Learning, Natural Language Processing, Supervised Machine Learning, Unsupervised Machine Learning, Algorithms, Cloud Computing (in progress), Data Science Capstone (in progress)
University of Washington, Seattle, WA — Sept. 2011 – Mar. 2015
Bachelor of Arts in Business Administration — Concentrations: Marketing, Information Systems

TECHNICAL KNOWLEDGE

Programming Languages:	Python, SQL
Cloud & Pipeline Orchestration:	Google Cloud Platform (Batch), Nextflow, Seqera/Tower
Libraries:	PyTorch, Pandas, NumPy, Scikit-learn, Matplotlib, SciPy
Machine Learning:	Deep learning architectures (U-Net, CNNs), model evaluation, statistical analysis
Tools:	Excel, Tableau, Jupyter

PROFESSIONAL EXPERIENCE

Atrium (contracted to GSK), San Jose, CA — June 2026 – Sept. 2026
Data Platform Engineering Intern
- Shipped 10+ features for Omics Workbench's Onyx Workflow Builder UI, GSK's self-service bioinformatics platform running 35+ Nextflow pipelines on Google Cloud Batch via Seqera Platform; opened 32 PRs across 2 production repos
- Built pipeline automation controls — retry-on-failure, retry-from-failed-node, and cancel/skip cascading — cutting manual intervention across large-scale multi-omics workflows
- Delivered dynamic pipeline configuration, subflow composition, and a Gantt/timeline view, improving setup speed and observability for complex workflows
- Built chat UI controls for a LangSmith-based LLM assistant (input debouncing, thread-stream abort), using a structured PRD-first AI-assisted development workflow to catch and correct off-spec AI-generated changes
- Took ownership of an underutilized Nextflow Tower metrics tool, transforming raw tabular pages into line, scatter, heatmap, and bubble charts and a summary home page for at-a-glance pipeline performance

Patagonia, Palo Alto, CA — Sept. 2025 – June 2026
Customer Experience Guide (Part-Time, alongside full-time graduate coursework)
Northeastern University, San Jose, CA — Jan. 2026 – May 2026
Teaching Assistant, Unsupervised Machine Learning
- Evaluated student assignments and provided technical feedback on clustering algorithms, dimensionality reduction, and anomaly detection techniques
- Held weekly office hours supporting 40+ graduate students with conceptual understanding and implementation challenges

Stanford Health Care (Blood Center), Palo Alto, CA — Feb. 2020 – Aug. 2024
Charge Technician (Dec. 2022 – Aug. 2024); Blood Center Technician I–III (Feb. 2020 – Dec. 2022)
- Led blood drive teams ensuring FDA compliance and operational efficiency, coordinating 50+ drives annually
- Self-initiated a donor retention project: identified early-donation drop-off, launched a pre-donation prep reminder in the donor app, and measured a ~15% lift in repeat-donation rate vs. a 1-year baseline in Excel; presented results to leadership

Foursquare (formerly Placed), Seattle, WA — May 2015 – Apr. 2017
App Marketing Specialist
- Analyzed app performance metrics using SQL across 10M+ user events, delivering insights that improved user engagement by 25%
- Designed and maintained 5+ Tableau dashboards tracking KPIs for internal stakeholders and affiliate partners

ACADEMIC PROJECTS

Few-Shot Brain Tumor Segmentation with Meta-Learning, CS 7150 Deep Learning, Northeastern University — Jan. 2026 – Apr. 2026
Python, PyTorch
- Built a ResNet34 U-Net baseline (24.4M parameters) and full BraTS 2020 preprocessing pipeline (369 patients, FLAIR/T1CE modalities) achieving 0.737 Dice under full supervision
- Implemented a prototypical-network few-shot segmentation method with a novel gated attention mechanism blending prototype similarity with U-Net output, reaching 0.829 mean Dice at k=20 support examples — exceeding the full-supervision baseline
- Designed cross-tumor transfer evaluation on the BraTS-PEDs pediatric dataset (257 patients, 200 episodes per k): meta-learning methods retained ~53% of in-distribution performance, versus ~77% degradation for baseline fine-tuning
- Integrated synthetic tumor generation (ellipsoid, spiculated, multifocal morphologies) for support-set augmentation, mixed at a 30% ratio with real training data

NYC Hospital Operational Analysis, Northeastern University, San Jose, CA — Jan. 2025 – Apr. 2025
Python, Scikit-learn
- Analyzed 2.3M+ patient records from the NY SPARCS dataset to identify cost drivers and care disparities across NYC boroughs
- Engineered features (log-transformed costs, severity categories) and trained Random Forest models to predict Length of Stay (R^2=0.72) and Mortality Risk (F1=0.85)
- Surfaced borough-specific insights: 22% higher sepsis treatment costs in Staten Island and 15% longer emergency stays in the Bronx
- Created 15+ interactive visualizations (heatmaps, joint plots) revealing key trends in patient demographics, costs, and clinical outcomes